1 Question (25 points)

Design a single tape TM that recognizes the language $L = \{0^n 1^p 2^q | p + q \text{ is a multiple of } n\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

2 Question (25 points)

Let $A_n = \{a^p b^q c^r |$ where $p + q + r$ is a multiple of $n\}$. Show that for each $n \geq 1$, the language is A_n is regular.

3 Question (25 points)

An all-NFA M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ that accepts $x \in \Sigma^*$, if every possible state that M could be in after reading input x is a state from F. Note in contrast, that an ordinary NFA accepts a string if some state among these possible states is an accept state. Prove that all-NFAs recognize the class of regular languages.

4 Question (25 points)

Let a k-PDA be a pushdown automaton that has k stacks. Thus a 0-PDA is an NFA, 1-PDA is conventional PDA. Give an example language for which 2-PDA recognizes that language but 1-PDA can not, prove it.